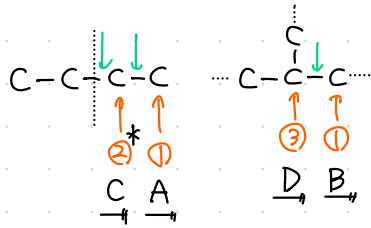
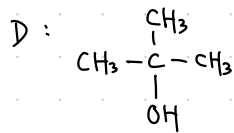
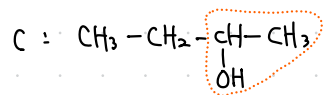
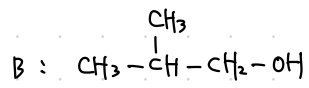
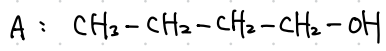


3-11 $C_4H_{10}O$ $U=1$ $\left\{ \begin{array}{l} \text{アルコール} \\ \text{エーテル} \end{array} \right.$



アルコール $A^{\oplus} B^{\oplus} C^* D^{\ominus}$
 O_x $\underline{E} \quad \underline{F} \quad G \quad X$
 $\text{アルコール} \quad \text{アルコール}$
 b.p. $A > B$

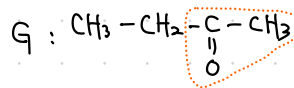
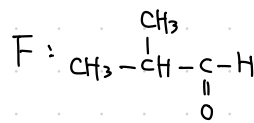
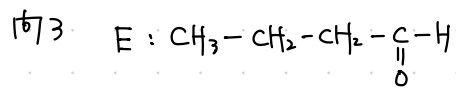
問1



問2

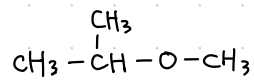
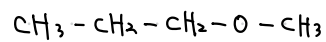
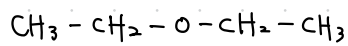
ヒドロキシ基を持ち、
分子間での水素結合を形成するため。

問3

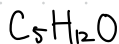


C, G

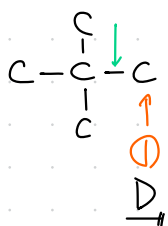
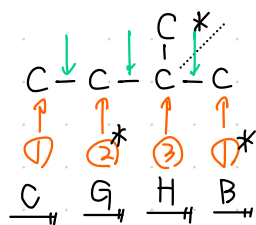
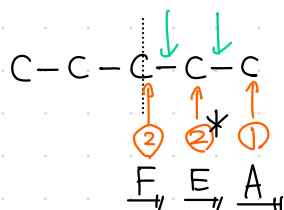
問4



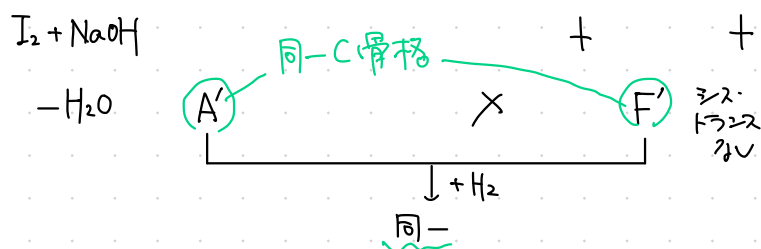
3 - 2



$$U=0 \quad \left\{ \begin{array}{l} \gamma_{\mu\nu} = -\gamma_{\nu\mu} \\ I = -\bar{I} \end{array} \right.$$

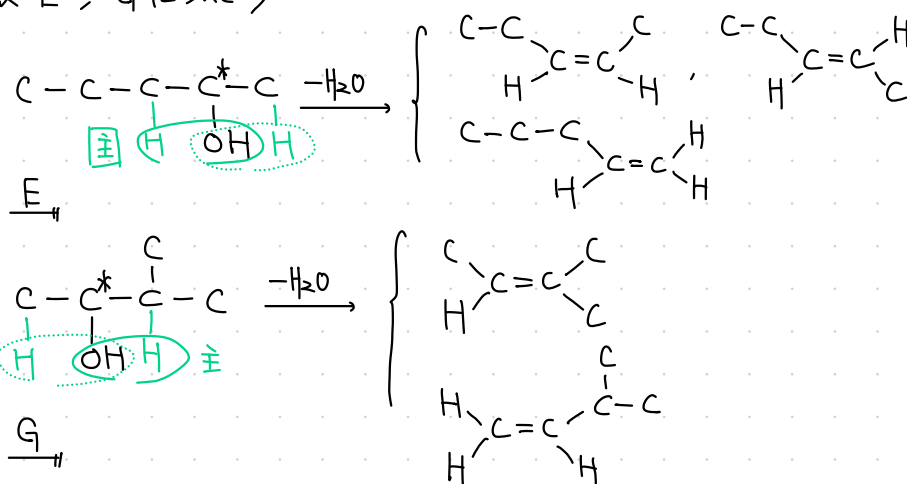


アルコール ① A ① B* ① C ① D ② E* ② F ② G* ③ H
Ox { アルデヒド } { ケトン } X



$\left. \begin{array}{l} \text{「脱水 } \sim \sim \text{」} \\ \text{「H}_2\text{付加 } \sim \sim \text{」} \end{array} \right\} \text{同一化合物}$
 $\longrightarrow \text{同-C骨格}$

$\langle 2 \text{級 } E^*, G^* | = \infty \rangle$

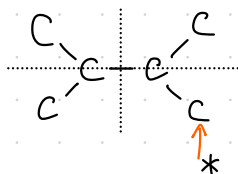
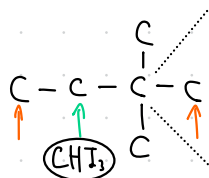
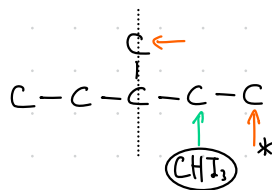
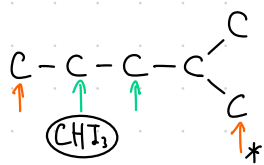
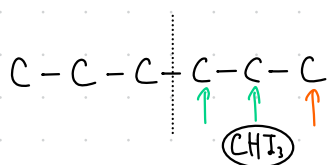


(2) $\frac{6}{4}$ (立体异构: 7)

3 - ③

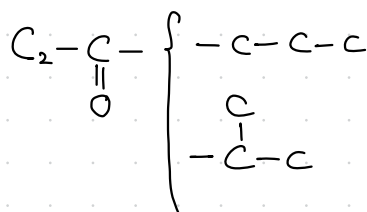
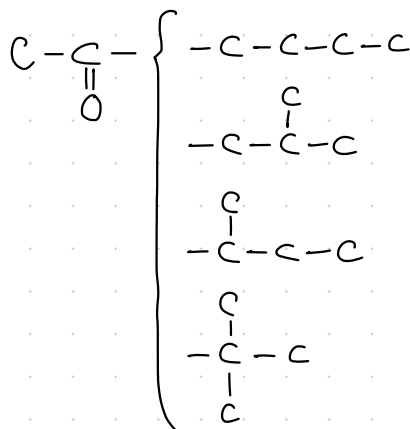
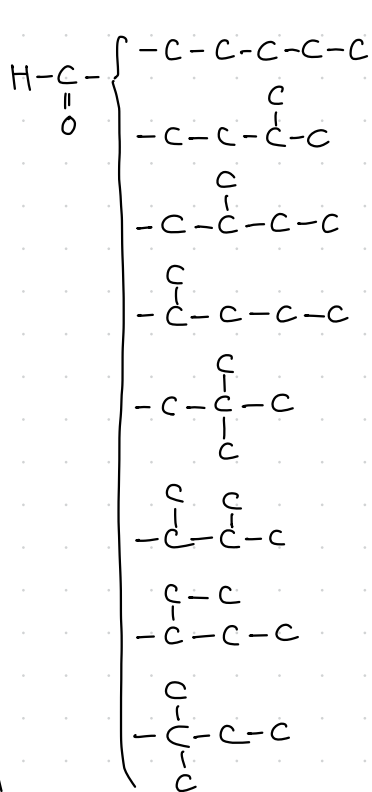
問1

← : =O (アルデヒド)
← : =O (ケトン)



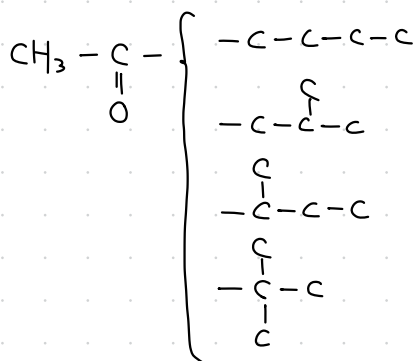
(a) 8

(b) 6



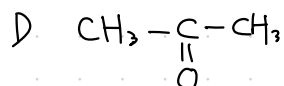
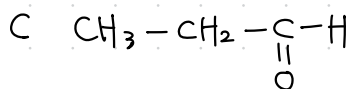
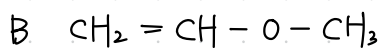
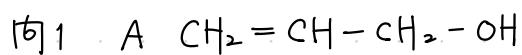
問2. 3

問3. 4

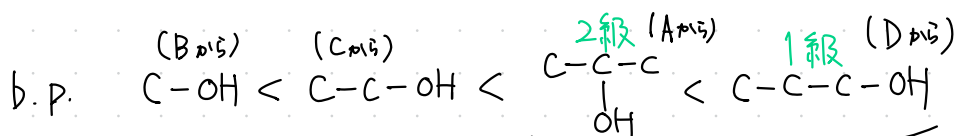
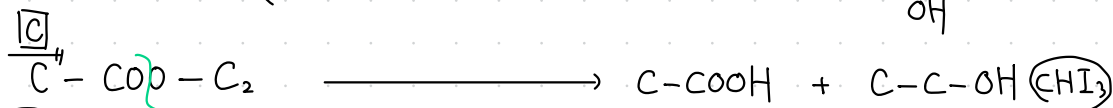
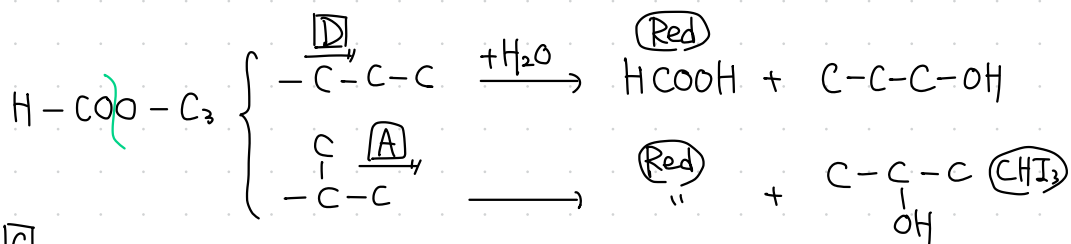
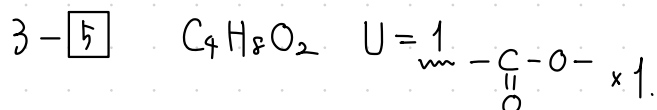
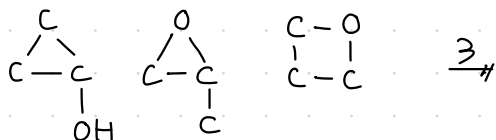


問4 X $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})-\text{H}$ Y $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})-\text{OH}$

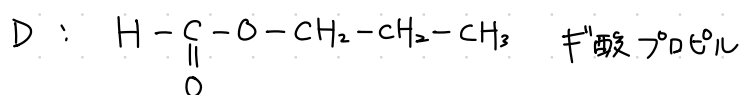
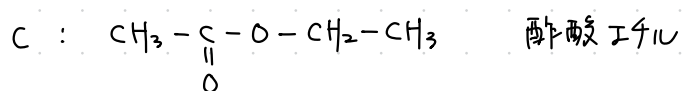
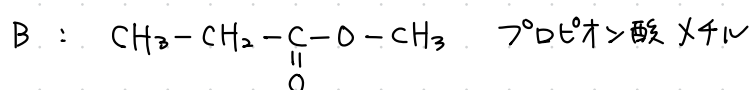
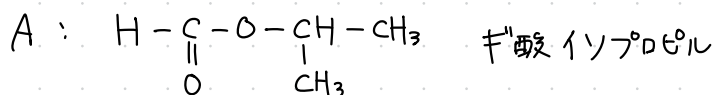
3-4



162



$(\text{小}) \leftarrow \text{分子量} \longrightarrow (\text{大}) \quad \text{異性体}$



3-6

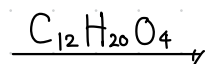
$$\text{例1} \quad w_C = 264 \text{ g} \times \frac{C(12)}{CO_2(44)} = 72 \text{ g}$$

$$w_H = 90 \text{ g} \times \frac{2H(2)}{H_2O(18)} = 10 \text{ g}$$

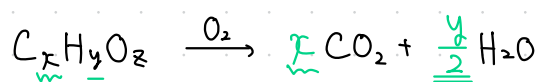
$$w_O = 114 - (72 + 10) = 32 \text{ g}$$

$$C : H : O = \frac{72 \text{ g}}{12 \text{ g/mol}} : \frac{10 \text{ g}}{1.0 \text{ g/mol}} : \frac{32 \text{ g}}{16 \text{ g/mol}} = 3 : 5 : 1$$

$$(C_3H_5O)_n = 57n = 228 \quad n = 4$$



<別解> 燃烧例, 分子量既知の時



$$C_xH_yO_z : CO_2 = 1 : x \text{ mol},$$

$$(288) \quad (44)$$

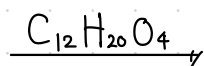
$$\frac{114 \text{ g}}{228 \text{ g/mol}} \times x = \frac{264 \text{ g}}{44 \text{ g/mol}} \times 1 \quad \therefore x = 12$$

$$C_xH_yO_z : H_2O = 1 : \frac{y}{2} \text{ mol}$$

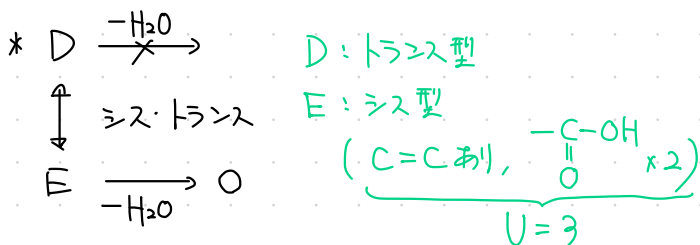
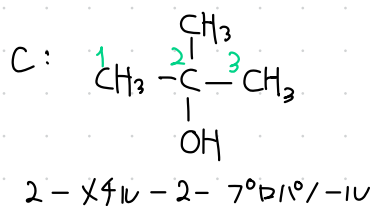
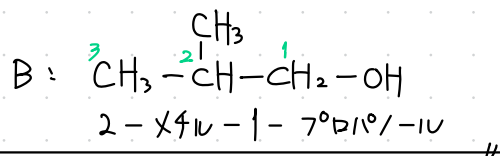
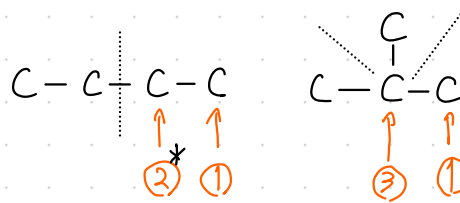
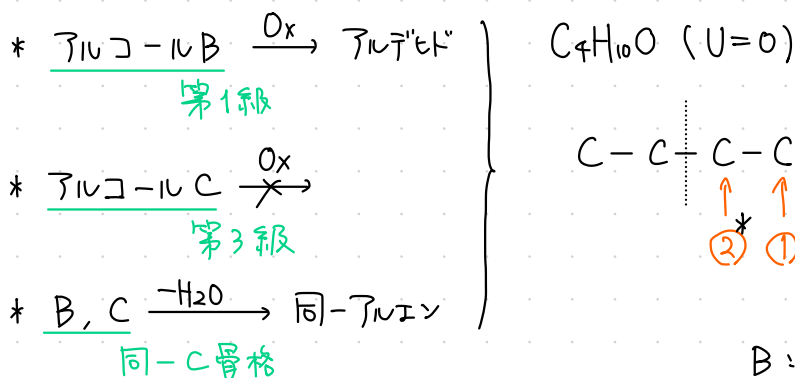
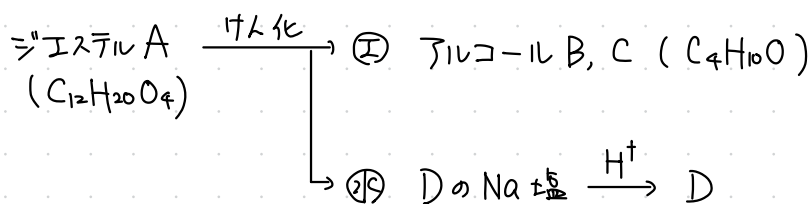
$$(288)$$

$$\frac{114 \text{ g}}{228 \text{ g/mol}} \times \frac{y}{2} = \frac{90 \text{ g}}{18 \text{ g/mol}} \times 1 \quad \therefore y = 20$$

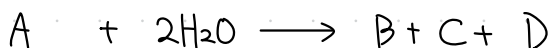
$$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4$$



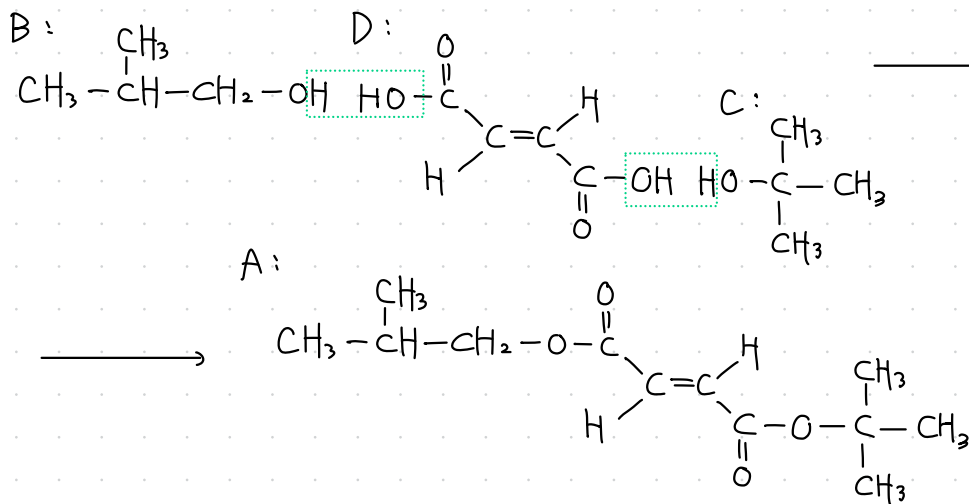
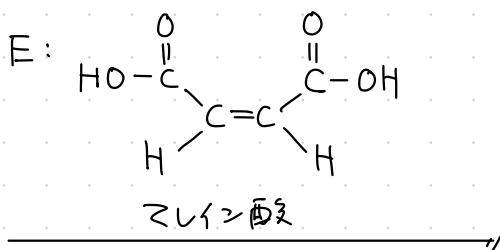
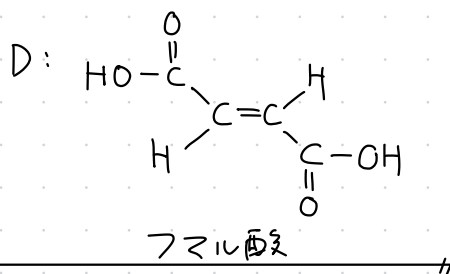
問2. A: $C_{12}H_{20}O_4$ $U = \frac{2 \times 12 + 2 - 20}{2} = 3$ C=C $\times 1$ + $-\overset{\overset{O}{\parallel}}{C}-O- \times 2$



ジエステル



$\therefore D = A + 2H_2O - (B + C)$
 $= C_{12}H_{20}O_4 + 2H_2O - 2 \times C_4H_{10}O$
 $= C_4H_4O_4$ (116)
 ($U=3$)



3 - 7

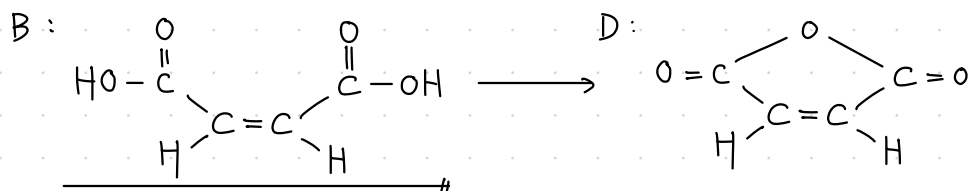
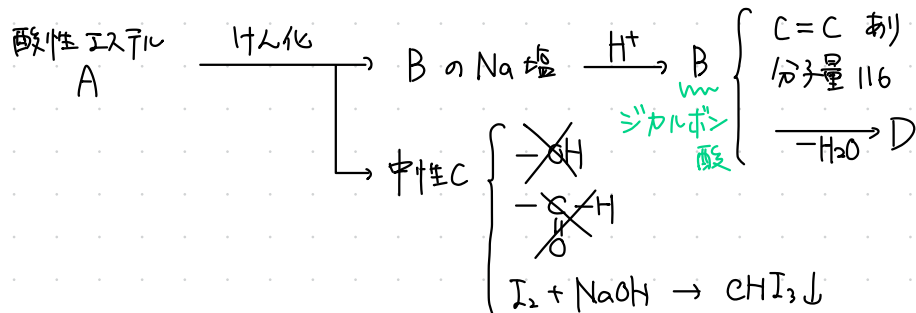
問1 $C_7H_8O_4$

$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \frac{2.56}{\text{①}}$$

$$\div 1.75 : 1.99 : 1 = 7:8:4$$

$$130 \leq (C_7H_8O_4)_n \leq 170 \quad \therefore n=1 \quad \underline{C_7H_8O_4} \quad (U=4)$$

問2

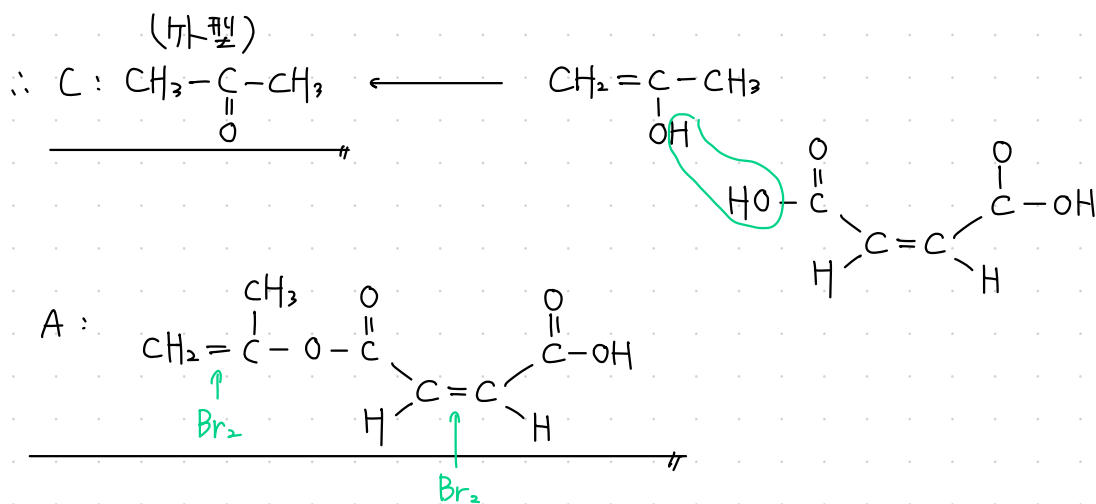


$$A + H_2O \longrightarrow B + C \quad \therefore C = A + H_2O - B$$

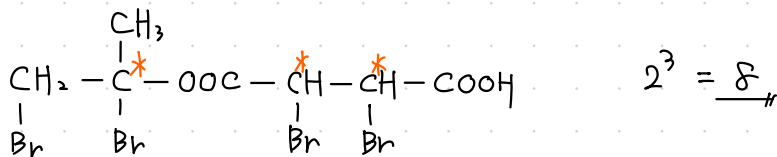
$$= C_7H_8O_4 + H_2O - C_4H_4O_4$$

$$= \underline{C_3H_6O} \quad (U=1)$$

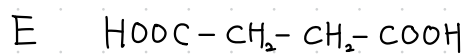
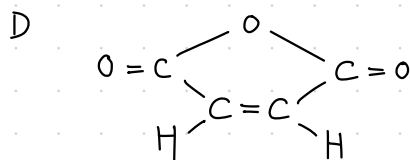
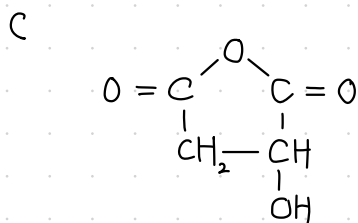
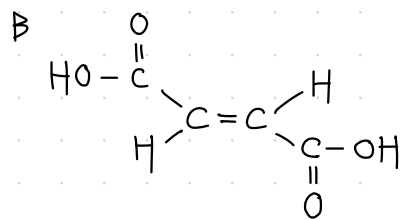
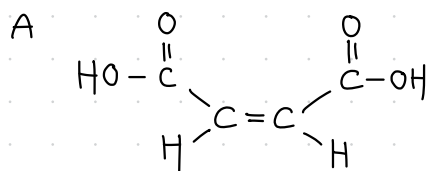
エステルの加水分解生酸性物として $-OH$ 基を持つものがない
 $\rightarrow$ エノール型 か ケト型 に変わった。



問3



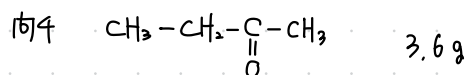
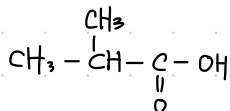
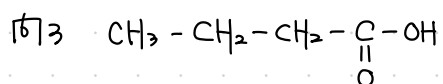
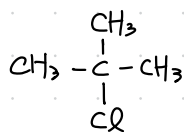
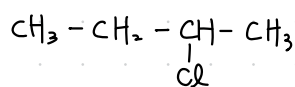
3-8



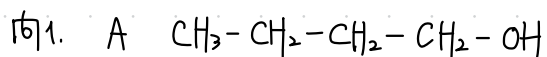
3-9

1 4

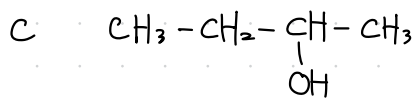
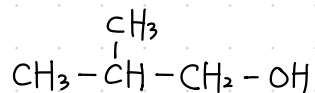
2.



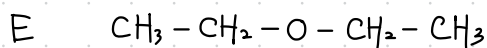
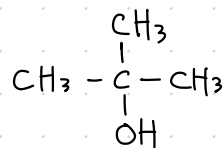
3- [10]



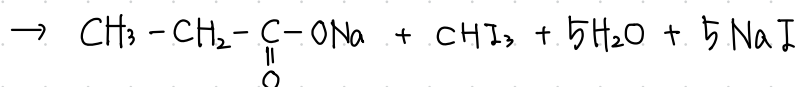
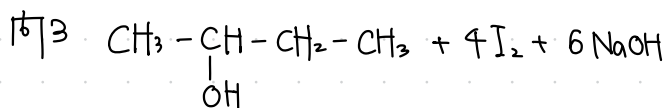
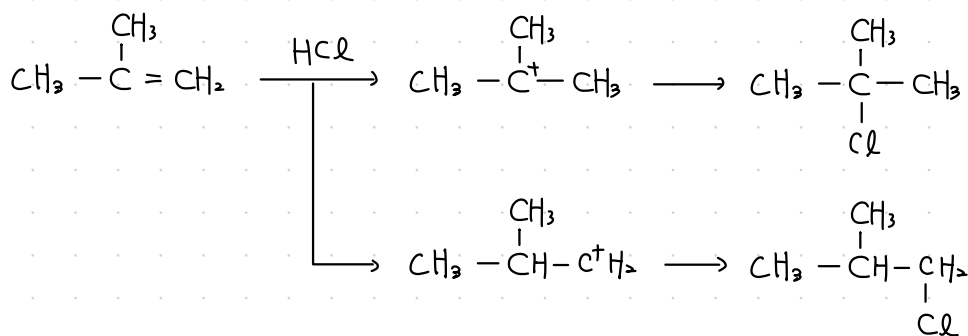
B



D



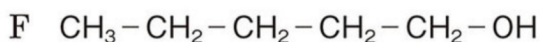
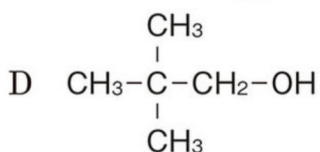
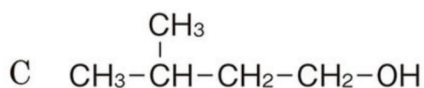
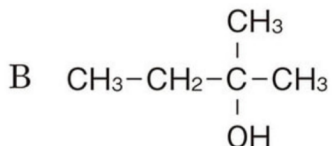
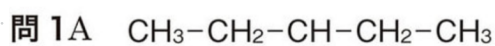
問2



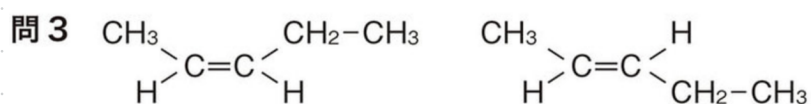
問4 C

問5 エ-テリ結合

3- [11]



問2 K



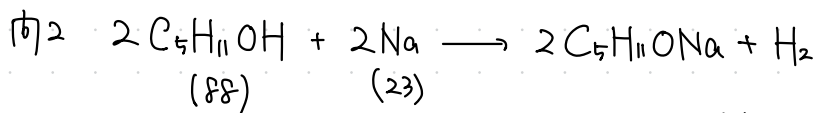
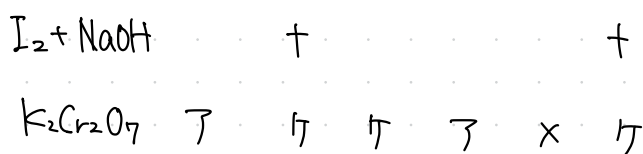
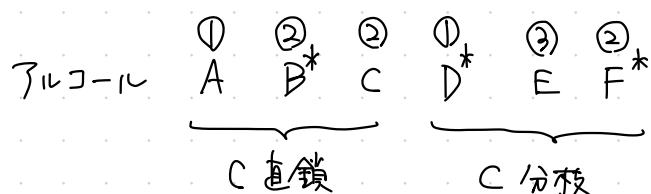
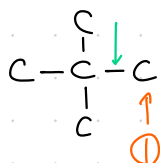
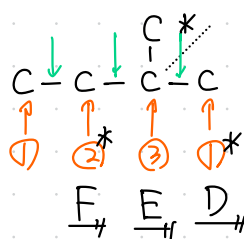
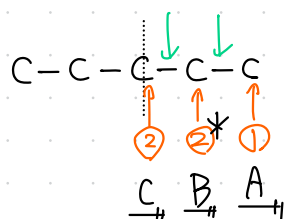
問1 A $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$
 B $\text{CH}_3-\text{CH}_2-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\text{C}}}-\text{CH}_3$
 C $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_3$ D $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\overset{\text{CH}_3}{\text{C}}}}-\text{CH}_2$
 E $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\overset{\text{CH}_3}{\text{C}}}}-\text{CH}_3$ F $\text{CH}_3-\underset{\text{OH}}{\underset{|}{\overset{\text{CH}_3}{\text{C}}}}-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CH}_3$

問2 0.15 L 問3 銀鏡反応, ホルミル アルデヒド基

問4 $\text{CH}_3-\text{CH}_2-\underset{\text{H}}{\underset{|}{\text{C}}}=\underset{\text{H}}{\underset{|}{\text{C}}}-\text{CH}_3$ $\text{CH}_3-\text{CH}_2-\underset{\text{H}}{\underset{|}{\text{C}}}=\underset{\text{CH}_3}{\underset{|}{\text{C}}}-\text{H}$
 $\text{CH}_3-\text{CH}_2-\text{CH}_2-\underset{\text{H}}{\underset{|}{\text{C}}}=\underset{\text{H}}{\underset{|}{\text{C}}}-\text{H}$

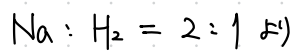
問5 $\text{H}_2\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{H}$ $\text{H}_2\text{C}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{H}$ $\text{H}_3\text{C}-\underset{\text{CH}_3}{\underset{|}{\text{C}}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{H}$
 $\text{H}_2\text{C}-\underset{\text{CH}-\text{CH}_2-\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{H}$ $\text{H}_2\text{C}-\underset{\text{CH}-\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{H}$ $\text{H}_2\text{C}-\underset{\text{CH}-\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{H}$

問6 $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\underset{|}{\overset{\text{CH}_3}{\text{C}}}}-\underset{\text{Cl}}{\underset{|}{\text{CH}_2}}$ $\text{CH}_3-\underset{\text{Cl}}{\underset{|}{\text{CH}}}-\underset{\text{OH}}{\underset{|}{\text{C}}}-\text{CH}_3$



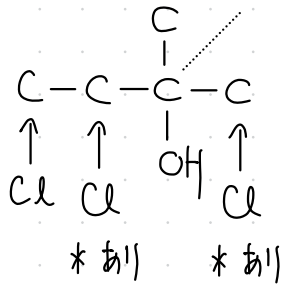
アルコール: $\frac{10\text{g}}{88\text{g/mol}} = \frac{5}{44}\text{mol}$
 Na: $\frac{0.3\text{g}}{23\text{g/mol}} = \frac{3}{230}\text{mol}$

係数も3:2なので、
 アルコール過剰出
 Naは合計で計算



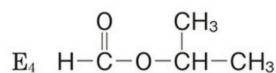
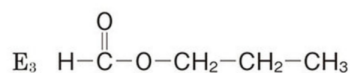
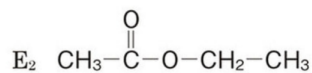
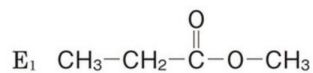
$$\frac{0.3}{230} \text{ mol} \times \frac{1}{2} \times 22.4 \text{ L/mol} = \frac{0.146}{5} \text{ L}$$

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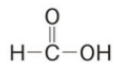


3-13

問1



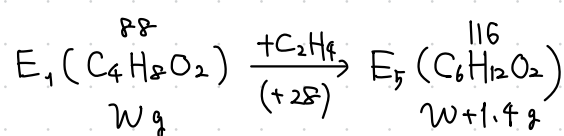
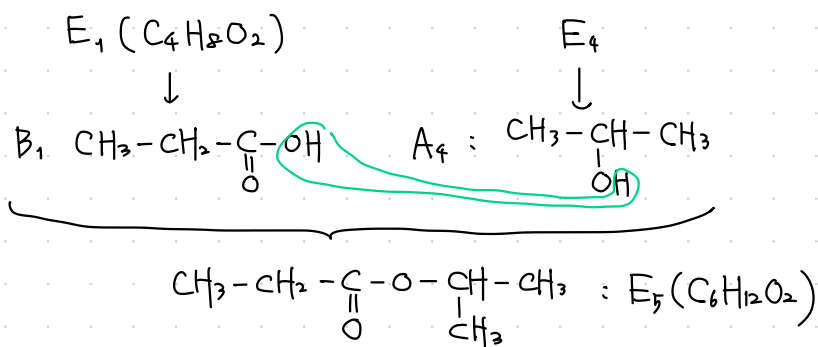
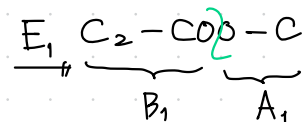
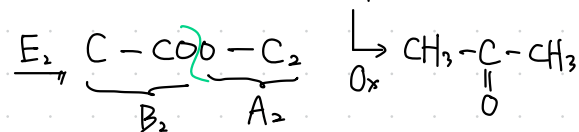
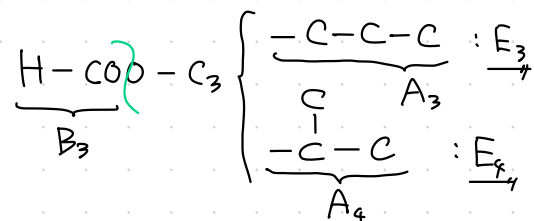
問2 ギ酸は分子内にホルミル基を持つため。(19字)



問3 酢酸カルシウム



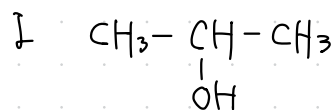
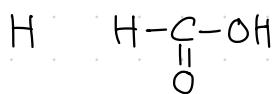
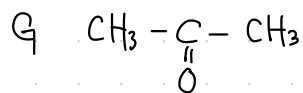
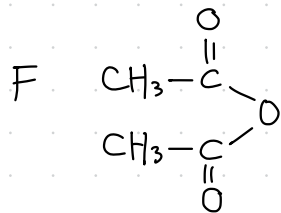
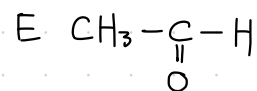
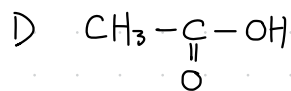
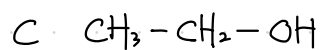
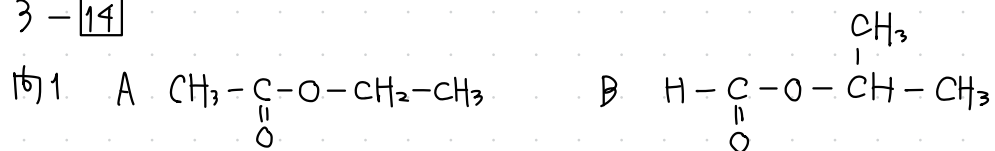
問4 4.4 g



$\frac{w \text{ g}}{88 \text{ g/mol}} = \frac{w+1.4 \text{ g}}{116 \text{ g/mol}}$

$\therefore w = 4.4 \text{ g}$

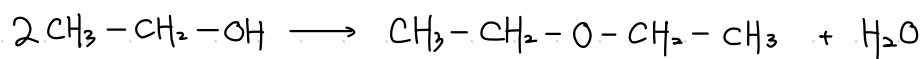
3-14

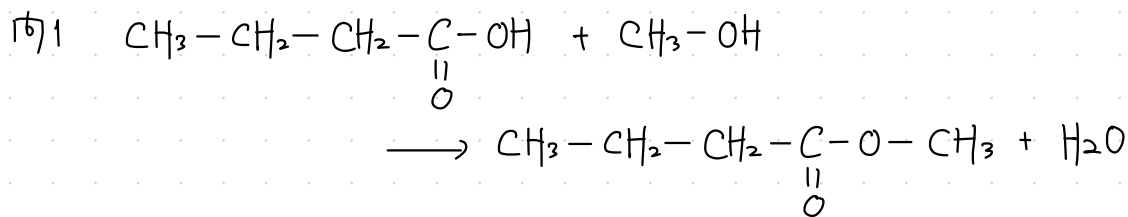


問2,3 略

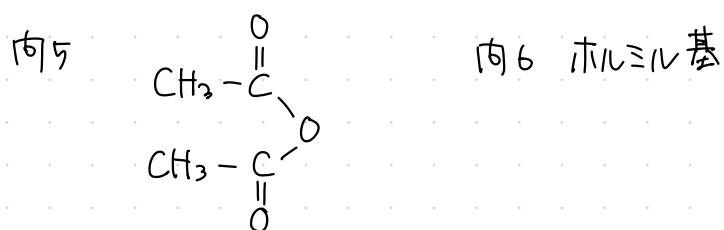
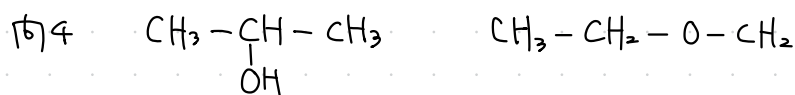
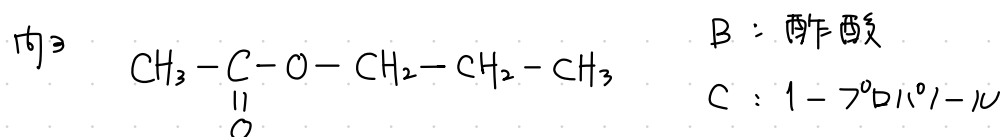
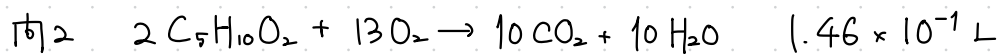
問4. テトラアセテート

問5. シエチルエーテル



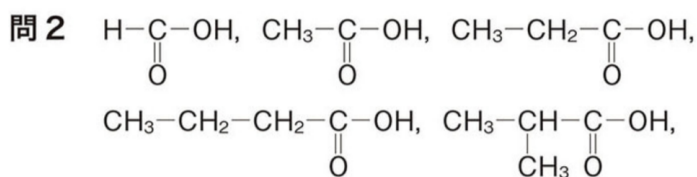
$3 - \boxed{15}$ 

エステル化の触媒

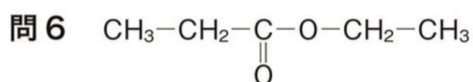
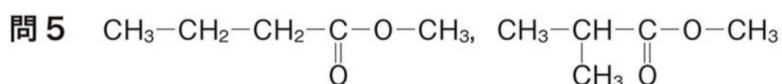
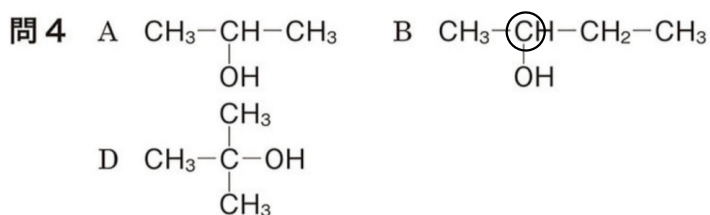


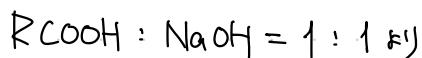
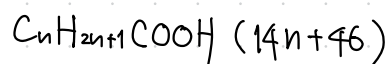
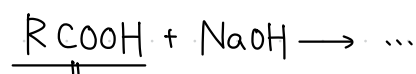
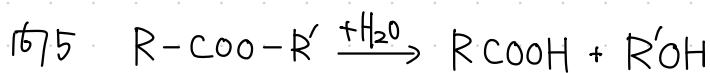
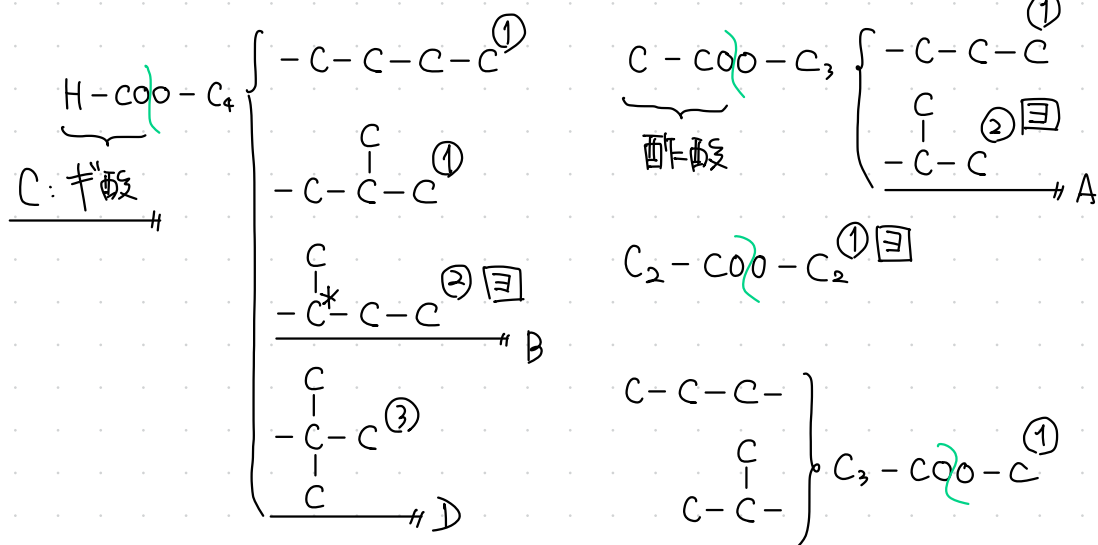
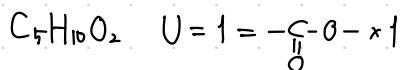
3 - 16

問 1 10

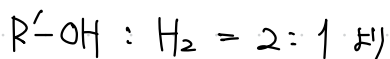
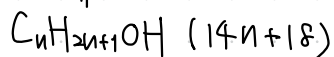
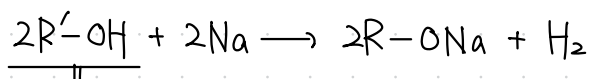
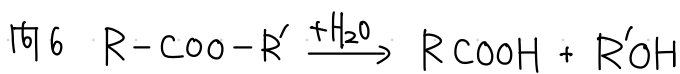
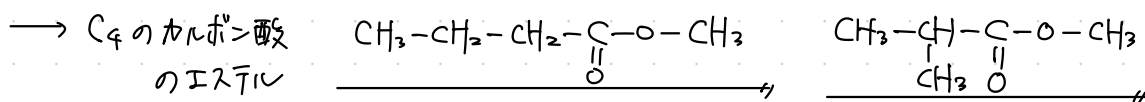


問 3 辛酸





$$\frac{528 \times 10^{-3} \text{ g}}{14n+46 \text{ g/mol}} = 0.10 \text{ mol/L} \times \frac{60}{1000} \text{ L} \quad \therefore n=3$$



$$\frac{230 \times 10^{-3} \text{ g}}{14n+18 \text{ g/mol}} \times 1 = \frac{56 \times 10^{-3} \text{ L}}{22.4 \text{ L/mol}} \times 2$$

